

RESEARCH DATASET REPORT

SEMICON v2 Dual Dataset

Complete Audit and Training-Readiness Assessment

Semiconductor MEMS Image Restoration and 2x Super-Resolution

4,158	97/100	92/100	95/100
PAIRS / TASK	SR QUALITY	DENOISING	HACKATHON

Dataset source: NFFA-Europe MEMS_devices_and_electrodes

Audit runtime: 7.66 minutes | Audit environment: Kaggle Python 3.12.13

Prepared for mentor review, Kaggle publication, and SEMICON India Hackathon 2026

Executive Summary

SEMICON v2 is a dual-purpose paired-image resource derived from 4,158 real NFFA-Europe MEMS scanning electron microscopy images. It provides a controlled 2x super-resolution benchmark and a same-resolution restoration benchmark while maintaining one deterministic source-level train/validation/test split across both tasks.

AUDIT VERDICT

Training-ready with documented cautions. Every referenced HR/LR pair exists, decodes successfully, follows the expected geometry, and has a matching sample identity. The principal research risks are 49 exact duplicate source groups, synthetic-to-real domain shift, fixed resizing, visible SEM annotation bars, and stale per-image metadata for the strengthened single-noise update.

Audit dimension	Result	Assessment
Super-resolution dataset	97 / 100	Excellent integrity and clean 2x protocol
Denoising dataset	92 / 100	Strong pairing; metadata provenance requires repair
Hackathon use	95 / 100	Highly useful external pretraining resource
Pair integrity	8,316 / 8,316 pairs valid	PASS
Corruption / missing / orphan files	0 / 0 / 0	PASS
Exact duplicate HR groups	49 groups; 103 files	WARNING

Key conclusions

- The SR subset is a scientifically clean bicubic-only 2x benchmark: 1024x768 HR maps exactly to 512x384 LR.
- The Noise subset contains 2,080 single-degradation and 2,078 combined-degradation pairs, with HR and noisy images both at 1024x768.
- The 70/20/10 split is exact: 2,910 train, 831 validation, and 417 test samples in each task.
- All 11 declared degradation operators are represented; single-type counts range from 169 to 213.
- Five strengthened single-noise groups were regenerated while the original manifest remained byte-identical, so the published per-image parameters are stale unless a repair manifest is supplied.

1. Dataset Purpose and Research Context

The dataset supports two complementary restoration problems encountered in semiconductor inspection. SEMICON-SR-2X-v2 isolates spatial-resolution recovery. SEMICON-Noise-v2 isolates artifact suppression and intensity restoration. Keeping these problems separate permits fair ablations, task-specific model selection, and a controlled sequential pipeline in which denoising and super-resolution are evaluated independently before integration.

1.1 Intended uses

- External pretraining and controlled benchmarking for SEM image super-resolution.
- Training and ablation studies for same-resolution SEM denoising and restoration.
- Comparison of CNN, transformer, hybrid, and GAN restoration architectures.
- Hackathon prototyping before final adaptation to the official KLA domain.

1.2 Out-of-scope interpretations

- The synthetic LR/noisy images are not direct measurements from a calibrated KLA acquisition system.
- Scores on this dataset should not be presented as official competition performance.
- Generated pixel detail must not be interpreted as metrologically valid semiconductor evidence without domain validation.

2. Dataset Construction and Shared Protocol

All 4,158 source images were converted to grayscale and standardized to 1024x768 before task generation. A fixed seed (2026) produced one shared image-level split. No patch-level split was used, so each source identity belongs to only one of train, validation, or test.

Split	Samples	Percentage	Noise single	Noise combined
Train	2,910	69.99%	1,455	1,455
Validation	831	19.99%	416	415
Test	417	10.03%	209	208
Total	4,158	100%	2,080	2,078

2.1 Folder organization

SEMICON-SR-2X-v2 stores HR and LR folders directly beneath each split. SEMICON-Noise-v2 nests single and combined categories inside every split, each containing HR and LR folders. Both manifests contain sample IDs, split assignments, source provenance, relative paths, dimensions, degradation type, and degradation parameters.

2.2 Storage profile

Subset	Pairs	Image files	Format	Disk size
SEMICON-SR-2X-v2	4,158	8,316	PNG	2.67 GiB
SEMICON-Noise-v2	4,158	8,316	PNG	4.23 GiB
Combined	8,316	16,632	PNG	6.90 GiB

3. Audit Scope and Methodology

1. Locate each manifest by content rather than relying only on folder names.
2. Resolve every HR and LR/noisy path and decode every referenced image.
3. Verify sample identifiers, filename stems, split assignments, dimensions, and expected task geometry.
4. Enumerate unreferenced image files and duplicated manifest IDs.
5. Hash all 4,158 unique SR HR files with SHA-256 to detect exact source duplicates.
6. Analyze degradation categories, parameter ranges, and single/combined balance from the manifest.
7. Sample 400 pairs per task for mean intensity, standard deviation, entropy, robust contrast, and Laplacian sharpness.
8. Generate visual inspections, training recommendations, and a transparent penalty-based scorecard.

READ-ONLY GUARANTEE

The audit read files under /kaggle/input and wrote reports only to /kaggle/working. It did not alter manifests, images, folder structure, or split assignments.

4. Integrity and Geometry Results

Check	SR result	Noise result	Status
Manifest records	4,158	4,158	PASS
Missing HR	0	0	PASS
Missing LR/noisy	0	0	PASS
Unreadable pairs	0	0	PASS
Name mismatches	0	0	PASS
Geometry errors	0	0	PASS
Duplicate sample IDs	0	0	PASS
Orphan image files	0	0	PASS

4.1 Resolution verification

Role	Expected	Observed	Count	Assessment
SR HR	1024x768	1024x768	4,158	PASS
SR LR	512x384	512x384	4,158	PASS
Noise HR	1024x768	1024x768	4,158	PASS

Role	Expected	Observed	Count	Assessment
Noise noisy	1024x768	1024x768	4,158	PASS

Every SR pair satisfies $HR\ width = 2 \times LR\ width$ and $HR\ height = 2 \times LR\ height$. Every denoising pair has equal HR and noisy dimensions.

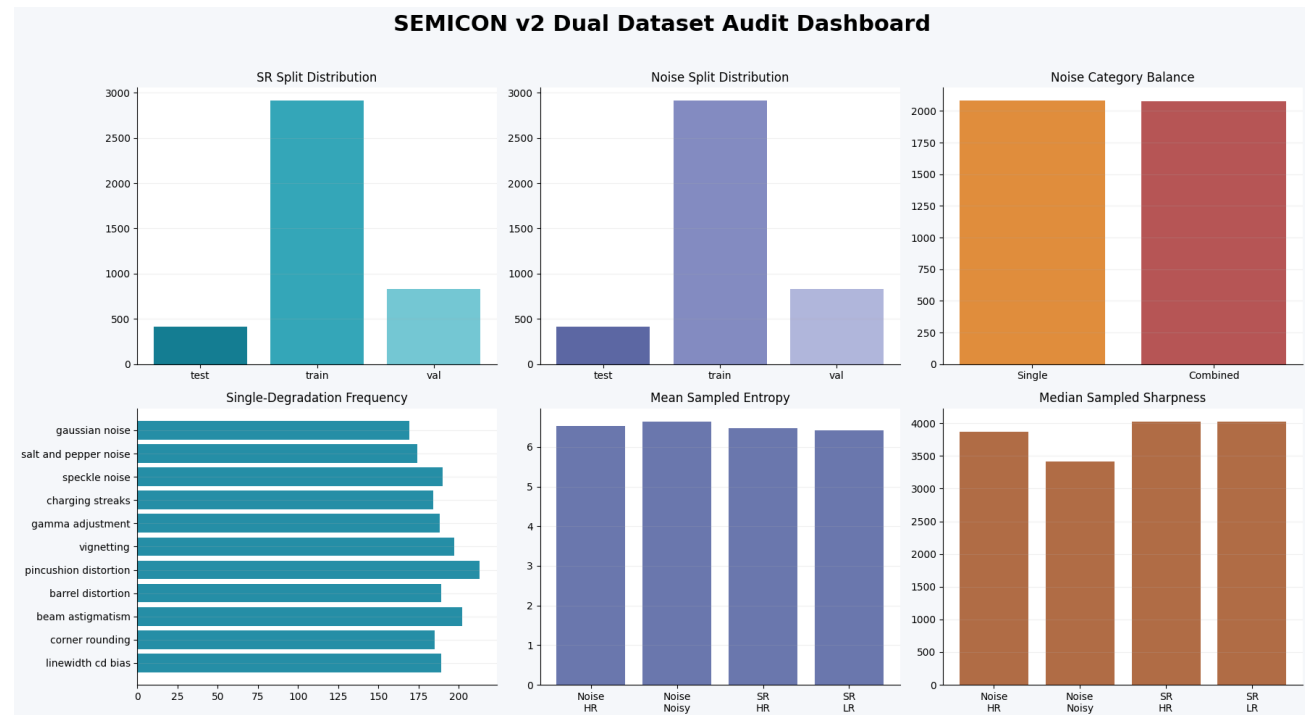


Figure 1. Executed audit dashboard: split balance, category balance, single-degradation frequency, entropy, and sharpness.

5. SEMICON-SR-2X-v2 Technical Assessment

The SR subset uses clean standardized HR images and bicubic 2x downsampling only. It contains no synthetic blur, noise, compression, or geometric degradation. This makes it appropriate for controlled reconstruction benchmarks and isolates architectural performance from degradation-model uncertainty.

Property	Specification
Task	Single-image 2x super-resolution
Input	512x384 grayscale PNG
Target	1024x768 grayscale PNG
Scale factor	Exactly 2 in width and height
Degradation	OpenCV INTER_CUBIC downsampling only
Pair count	4,158

Property	Specification
Audit score	97/100

5.1 Strengths

- Unambiguous scale geometry and complete pairing.
- No confounding restoration artifacts in the SR benchmark.
- Real SEM/MEMS structures provide domain-relevant edges, textures, and repeated patterns.
- Fixed dimensions simplify batching and reproducibility while training can still use random crops.

5.2 Limitations

- The original images were resized to 1024x768, potentially changing native physical scale.
- A single bicubic kernel does not cover unknown real-world degradation kernels.
- Scale bars and instrument text may dominate pixel metrics or become shortcuts.
- Exact source duplicates slightly inflate apparent diversity.

6. SEMICON-Noise-v2 Technical Assessment

The denoising subset contains one clean HR image and one same-resolution degraded image per source. Single samples teach identifiable operator-specific inverse mappings; combined samples expose the model to all 11 operators applied sequentially.

Property	Specification
Task	Same-resolution restoration / denoising
Input and target	1024x768 grayscale PNG
Single samples	2,080
Combined samples	2,078
Combined policy	All 11 operators
Audit score	92/100

6.1 Degradation inventory and strengths

Degradation	Parameter	Effective single range	Metadata source	Single N
Linewidth/CD bias	radius; erosion/dilation	1-3 px; random direction	Original manifest	189
Corner rounding	Gaussian radius	0.5025-1.9974 px	Original manifest	185
Beam astigmatism	sigma x / sigma y	0.3009-1.1995 / 0.3001-1.1999	Original manifest	202

Degradation	Parameter	Effective single range	Metadata source	Single N
Barrel distortion	k1	0.000294-0.119901	Original manifest	189
Pincushion distortion	k1	-0.119977 to -0.000110	Original manifest	213
Salt-and-pepper	probability	0.000101-0.009999	Original manifest	174
Charging streaks	intensity; count; half-width	0.10-0.28; 8-20; 3-10 px	Strong single override	184
Vignetting	strength	0.22-0.55	Strong single override	197
Speckle noise	sigma	0.08-0.20	Strong single override	190
Gaussian noise	sigma	0.035-0.095	Strong single override	169
Gamma adjustment	gamma	0.50-0.75 or 1.35-1.70	Strong single override	188

METADATA PROVENANCE WARNING

The strong-update notebook replaced selected single-noise LR images but intentionally preserved manifest.json byte-for-byte. The manifest still reports original ranges (for example gamma 0.8-1.2 and charging frequency 1-6). Exact updated per-image values require the separate update log or a verified repaired manifest.

6.2 Distribution balance

Single-degradation counts range from 169 Gaussian-noise samples to 213 pincushion-distortion samples. This is a modest imbalance rather than a severe skew. Combined samples contribute 2,078 appearances of every operator, but combined examples should not be counted as clean operator-isolation evidence because all effects co-occur.

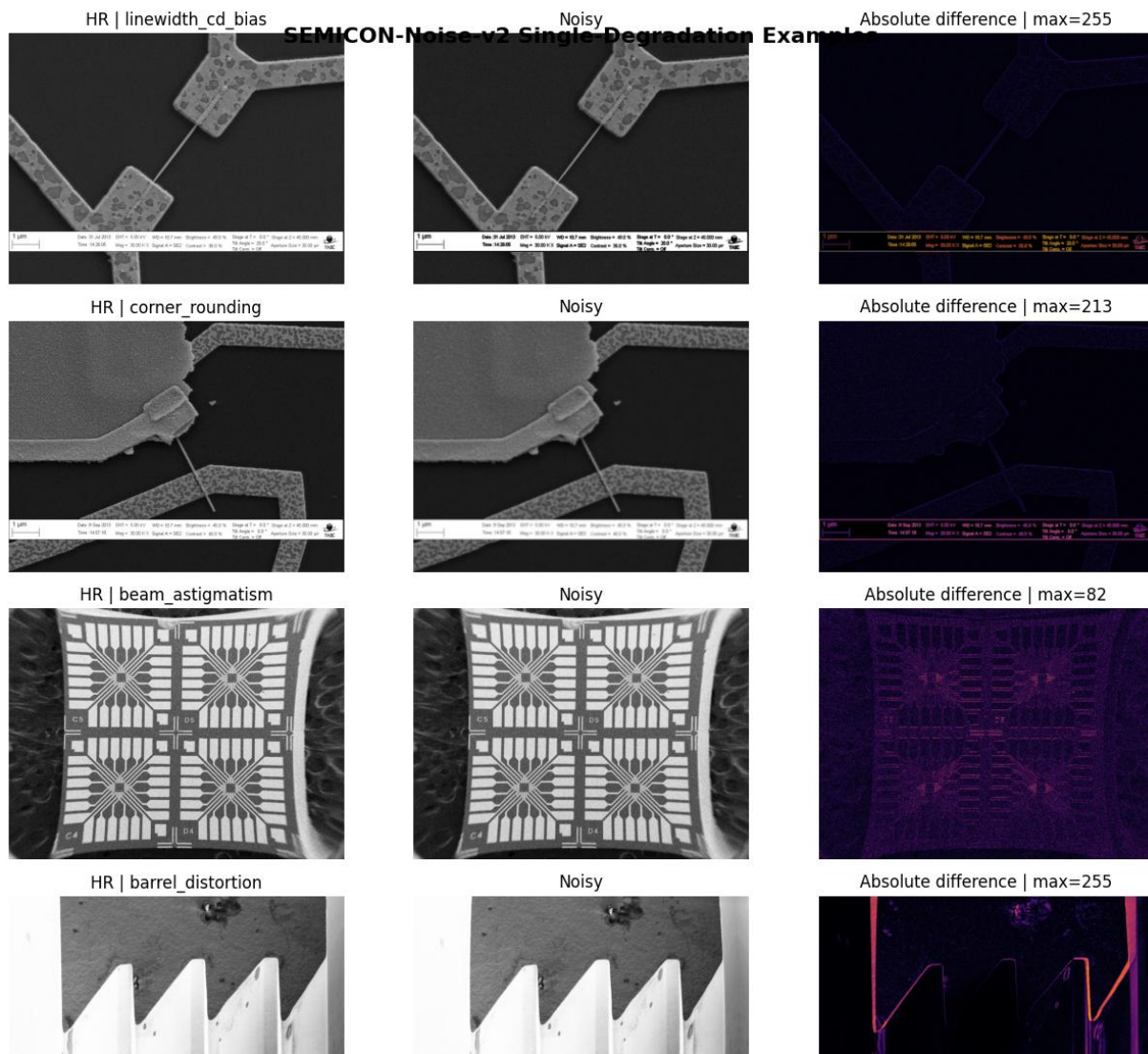
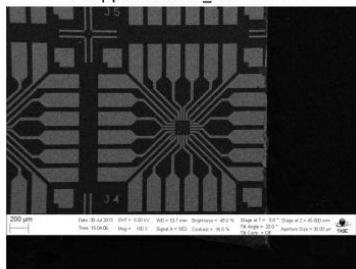


Figure 2a. Single-degradation inspection (part 1 of 3): clean HR, noisy image, and absolute difference map.



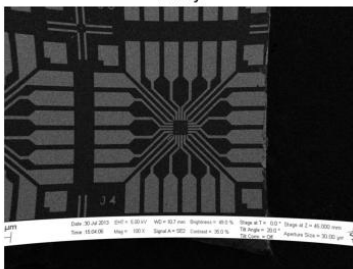
HR | pincushion_distortion



HR | vignetting



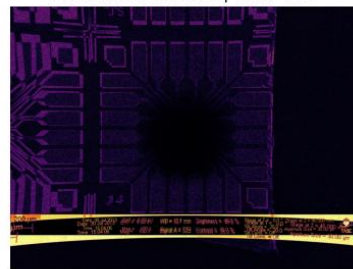
Noisy



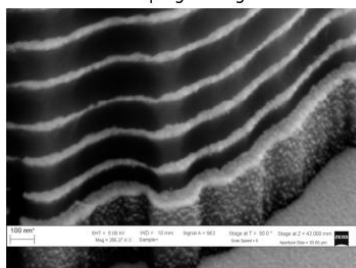
Noisy



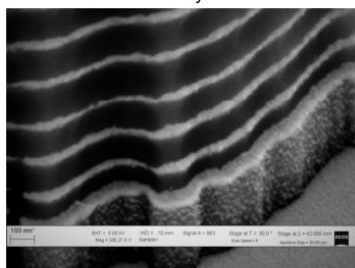
Absolute difference | max=255



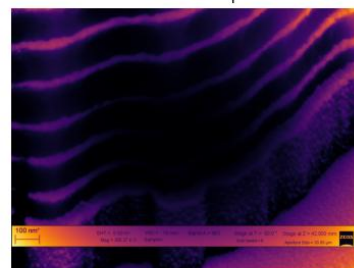
Absolute difference | max=87



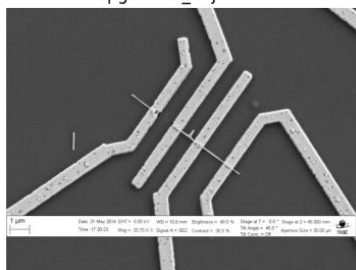
HR | gamma_adjustment



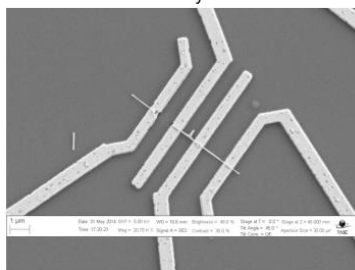
Noisy



Absolute difference | max=46



HR | charging_streaks



Noisy



Absolute difference | max=59

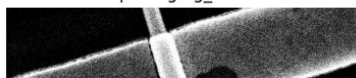


Figure 2b. Single-degradation inspection (part 2 of 3): clean HR, noisy image, and absolute difference map.

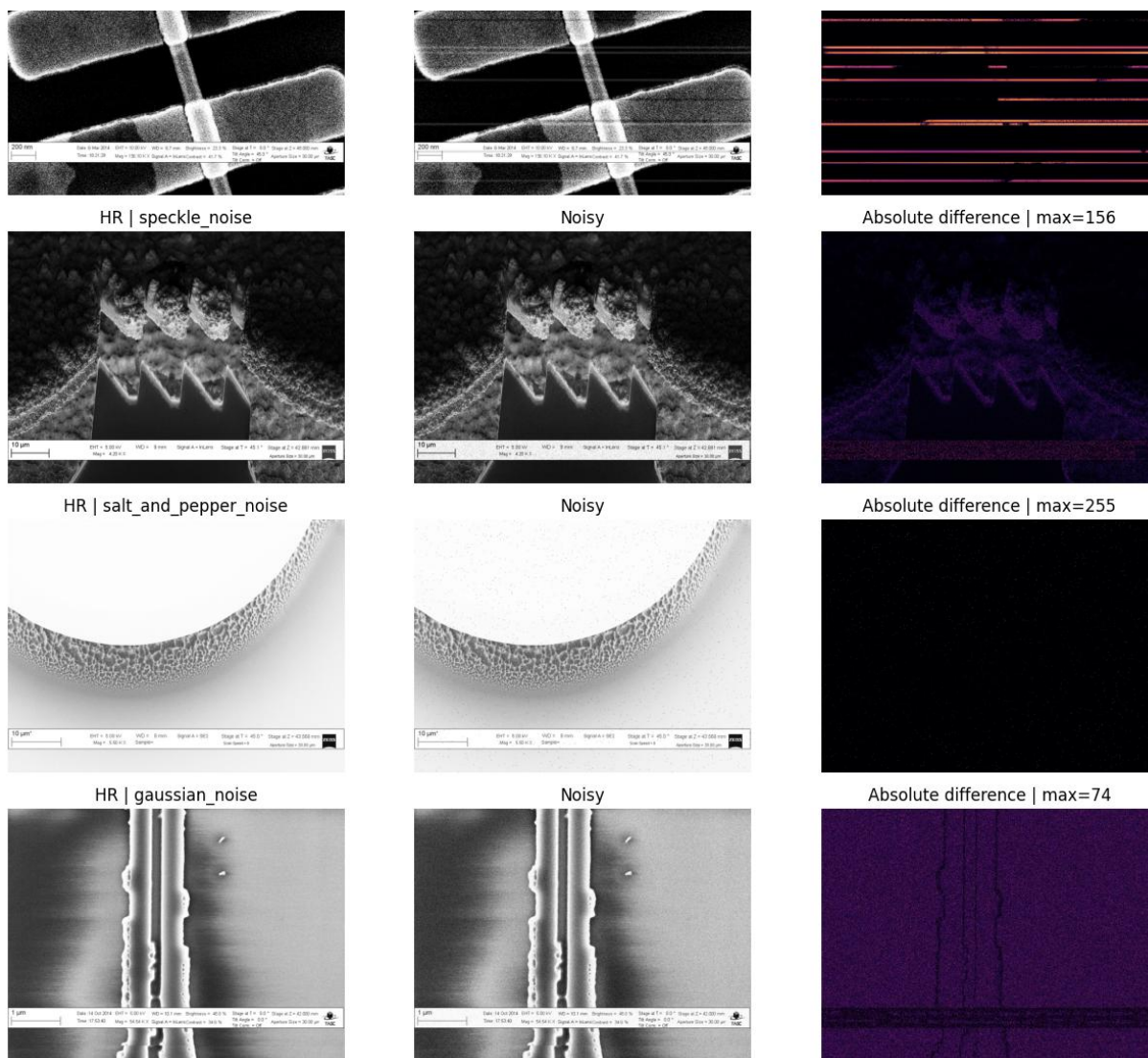


Figure 2c. Single-degradation inspection (part 3 of 3): clean HR, noisy image, and absolute difference map.

7. Sampled Image-Quality Analysis

A deterministic sample of 400 pairs per task was analyzed. Metrics describe data characteristics, not model performance. Sharpness is the variance of the Laplacian; contrast is the 95th minus 5th intensity percentile; entropy is computed from the 8-bit histogram.

Dataset role	Mean intensity	Intensity std	Entropy	Contrast	Sharpness
SR HR	0.43641	0.23953	6.47798	0.81854	5,353.23
SR LR	0.43572	0.23432	6.40958	0.81103	5,028.22
Noise HR	0.44148	0.24131	6.51922	0.82596	4,976.08
Noise noisy	0.42961	0.22415	6.63810	0.73341	4,052.79

7.1 Interpretation

- Bicubic downsampling slightly reduces entropy, contrast, and mean sharpness while preserving global intensity.
- Noise images show higher entropy but lower robust contrast and lower Laplacian sharpness than their HR counterparts, consistent with mixed blur/noise/intensity artifacts.
- Large standard deviations across samples reflect substantial variation in SEM structure and acquisition appearance.
- These descriptive statistics should be complemented by per-degradation PSNR/SSIM/MAE distributions when tuning severity or reporting model results.

8. Duplicate and Leakage Risk

WARNING: 49 EXACT DUPLICATE GROUPS

SHA-256 hashing identified 49 groups involving 103 HR files. Most groups contain two files; at least one group contains four. The audit confirms duplicate content, not merely similar dimensions or filenames.

The original split was image-level and therefore prevents crops from one filename crossing splits. It does not guarantee that byte-identical or near-duplicate source files remain together. If duplicate members occur across train, validation, and test, evaluation may be optimistic.

Recommended duplicate-control procedure

9. Compute exact hashes and perceptual hashes on the original source images.
10. Build connected duplicate and near-duplicate groups.
11. Assign every group atomically to one split.
12. Regenerate both task manifests from the same grouped split ledger.
13. Retain the current split for benchmark continuity, but disclose its duplicate limitation.

9. Training Readiness

9.1 Super-resolution configurations

Model	Crop	Batch (T4 16GB)	Epochs	Expected VRAM	Role
SRCNN	128-256	8	100	2-4 GiB	Fast sanity baseline
EDSR	192	4	150	5-7 GiB	Strong PSNR baseline
RCAN	128-192	2	200	7-10 GiB	Channel-attention benchmark
ESRGAN	128	2	150	8-12 GiB	Perceptual two-stage GAN
SwinIR	128	2	150	9-13 GiB	Window-transformer benchmark
HAT	96-128	1	150	12-16 GiB	High-cost final model

9.2 Denoising configurations

Model	Crop	Batch (T4 16GB)	Epochs	Expected VRAM	Role
Restormer	128-192	2	100	9-13 GiB	Strong global/local restoration
NAFNet	256	4	100	7-10 GiB	Efficient high-PSNR choice
MIRNet-v2	128	2	100	10-14 GiB	Multi-scale restoration
MPRNet	128-192	2	100	9-13 GiB	Multi-stage supervision
Pix2PixHD	128-256	1	100	10-14 GiB	Secondary GAN benchmark
BSRGAN	128	1	100	10-15 GiB	Reduced RRDB configuration

9.3 Recommended training protocol

- Use manifest-defined splits without resplitting or mixing task identities.
- Train on random aligned patches; validate and test with full-image or overlap-tiled inference.
- Use deterministic seeds, AMP, gradient clipping, checkpoint resume, and best-validation-PSNR selection.
- Report PSNR, SSIM, MAE, MSE, RMSE, runtime, parameters, and memory. Compute expensive no-reference metrics only during final evaluation.
- For Noise-v2, report aggregate, single, combined, and per-degradation results separately.

10. Strengths

- Real MEMS/SEM content supplies domain-relevant edges, electrodes, textures, cavities, and repetitive structures.
- A shared source-level split supports fair sequential and multi-task comparisons.
- Every pair is readable, correctly named, dimensionally valid, and represented in the manifest.
- Pure bicubic SR creates a clean controlled benchmark.
- The denoising task spans morphology, blur, geometric distortion, illumination, intensity response, structured charging, and stochastic noise.
- Balanced single/combined categories support operator-specific learning and multi-degradation robustness.

11. Weaknesses and Research Risks

Risk	Impact	Priority	Required mitigation
Synthetic-to-real gap	Models may not transfer to official KLA degradation	High	Fine-tune and validate on official data
Stale strong-noise metadata	Per-image provenance is incomplete	High	Publish update log or repaired manifest
Exact duplicate sources	Potential leakage and optimistic validation	High	Group duplicates before future split revision

Risk	Impact	Priority	Required mitigation
Fixed 1024x768 resizing	Native physical scale is altered	Medium	Use scale-aware crops and disclose preprocessing
All-11 combined policy	May be more severe than real acquisition	Medium	Add realistic subsets in future work
Annotation bars / scale labels	Shortcut learning and metric distortion	Medium	Mask, crop, or evaluate content ROI
Single bicubic SR kernel	Limited degradation diversity	Low for controlled benchmark	Keep as benchmark; add blind-SR data separately

12. Hackathon Strategy

12.1 Recommended model sequence

PREFERRED PIPELINE

Degraded SEM input -> denoising/restoration model -> 2x super-resolution model -> final range-safe output. This order removes stochastic and structured artifacts before upscaling, reducing the risk of magnifying noise and charging streaks.

The team should benchmark the reverse order as an ablation because the official competition degradation order may differ. Select the sequence using the official validation distribution rather than synthetic-set metrics alone.

12.2 Model priorities

- SR baseline: EDSR. Strong SR candidate: RCAN or lightweight SwinIR. Perceptual secondary model: ESRGAN.
- Denoising baseline: NAFNet. Strong global model: Restormer. Multi-scale alternative: MIRNet-v2 or MPRNet.
- Reserve HAT and large GANs for later experiments after stable baselines, runtime profiling, and metric validation.

12.3 Fair benchmark rules

- Use identical train/validation/test manifests across all models.
- Use the same crop policy, evaluation border handling, intensity range, and metric implementation.
- Evaluate only the best validation checkpoint on test data.
- Measure inference after warm-up with synchronized GPU timing and report image resolution.
- Do not compare perceptual GAN results solely by PSNR; include LPIPS and expert visual inspection where permitted.

13. Dataset Governance and Reproducibility

14. Publish the original manifest, corrected strong-noise metadata, shared split ledger, and generation notebooks together.
15. Record SHA-256 hashes for manifests and release archives.

16. Document the exact OpenCV, NumPy, SciPy, and Pillow versions used for generation.
17. Maintain an immutable release tag; do not silently replace images under the same version.
18. Provide a changelog describing image changes separately from metadata-only corrections.
19. Preserve NFFA-Europe source licensing, attribution, and redistribution requirements in the dataset card.

14. Final Scorecard and Verdict

Dimension	Score	Rationale
Super Resolution	97 / 100	Perfect pair integrity and geometry; minor duplicate/domain limitations
Denoising	92 / 100	Strong coverage and balance; stale updated metadata and synthetic gap
Hackathon Use	95 / 100	Excellent external pretraining resource with manageable cautions

FINAL VERDICT: APPROVED WITH DOCUMENTED WARNINGS

The dual dataset is suitable for model development, controlled benchmarking, and external pretraining for the SEMICON India Hackathon. It should not be treated as the final target-domain benchmark. Before publication is frozen, repair the strong single-noise metadata and disclose the exact duplicate groups. Before final submission, fine-tune and select models using official KLA-domain validation data.

15. Recommended Action Plan

Priority	Action	Owner outcome
P0	Publish corrected per-image strong-noise parameters	Restores reproducibility
P0	Audit duplicate-group split membership	Quantifies leakage risk
P1	Establish EDSR and NAFNet/Restormer baselines	Reliable reference results
P1	Standardize metrics and border conventions	Fair comparison
P1	Evaluate by noise category and degradation type	Failure-mode visibility
P2	Mask or crop annotation bars in a controlled ablation	Reduces shortcut learning
P2	Fine-tune on official KLA data	Target-domain adaptation

Appendix A. Manifest Requirements

- Common fields: sample_id, split, source_image, original dimensions, HR dimensions, hr_path, and lr_path.
- SR fields: scale_factor=2, degradation_type=bicubic_downsampling_only, interpolation metadata.
- Noise fields: category, seed, degradation_type, and exact per-image degradation_parameters.

- A metadata repair must not change sample IDs, splits, categories, paths, seeds, or image bytes.

Appendix B. Audit Artifacts

Artifact	Purpose
pair_audit.csv	Per-pair existence, readability, geometry, and naming results
duplicate_groups.csv	Exact SHA-256 duplicate groups
degradation_summary.csv	Operator frequencies and parameter ranges
sampled_image_quality.csv	Sample-level image statistics
quality_summary.csv	Aggregated descriptive quality statistics
dataset_audit_dashboard.png	Visual overview of splits, degradation balance, entropy, and sharpness
SEMICON_v2_DUAL_DATASET_AUDIT_REPORT.pdf	Automated audit summary generated by the Kaggle notebook

Appendix C. Source Basis

- Executed notebook: semicon-v2-dual-dataset-audit.ipynb.
- Dataset generator: SEMICON_Dual_Dataset_v2_Generator.ipynb.
- Strong single-noise update: SEMICON_Noise_v2_Strong_Single_Update_and_ZIP.ipynb.
- Dataset: SEMICON v2 Dual Dataset on Kaggle, containing SEMICON-SR-2X-v2 and SEMICON-Noise-v2.